USER GUIDE

IBEX Machine Learning Studio

Getting started and your first model

A practical guide to importing data, preparing features, training models and understanding predictions. Follow the on-screen control names in bold and use the illustrated checks to confirm each stage.

What you will do

Start from Home, import a CSV, inspect the dataset, choose appropriate preprocessing and train a linear regression model. You will also learn how to enter original feature values for prediction and save your results.

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The illustrated dataset

The supplied example contains 768 rows, eight input columns named X1 to X8 and one target named y1. These names are retained throughout the examples; use the meanings and units defined for your own dataset.

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01 GETTING STARTED

Find your way around the studio

Open the application to begin at Home. The three cards lead to Regression, Classification and Clustering. For a new analysis, first select **Dataset** in the left sidebar and import your CSV.

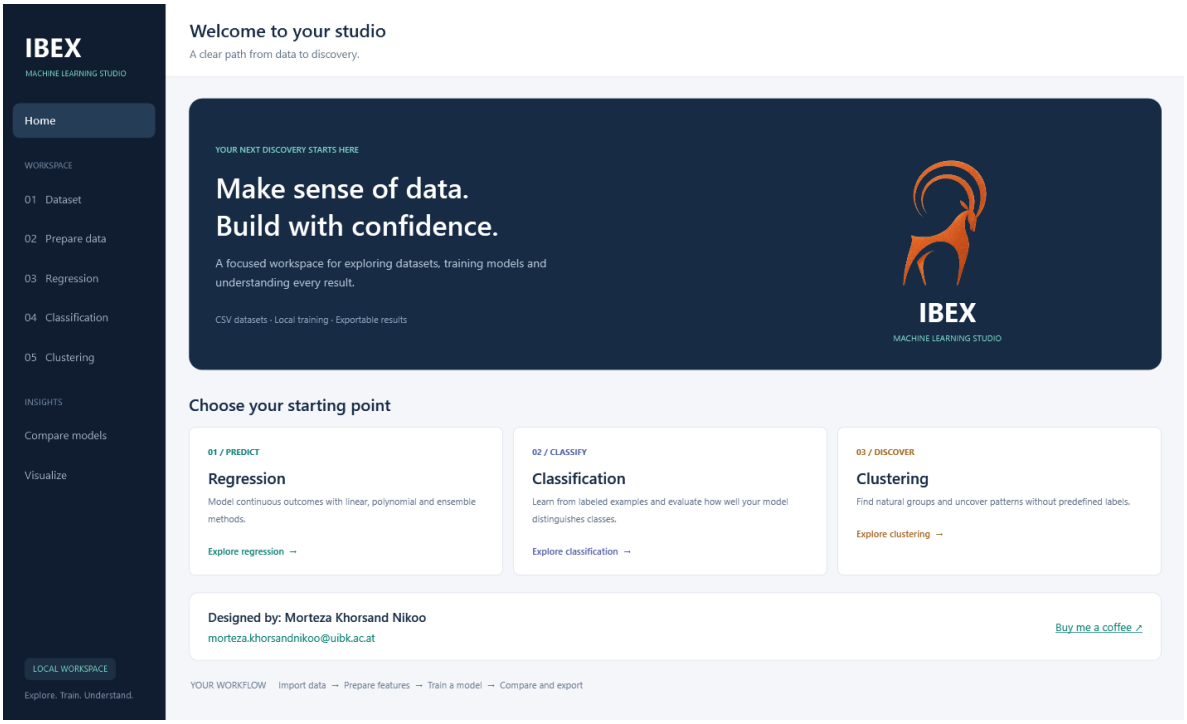


Figure 1 Home introduces the three modeling tasks. Use the sidebar to move between pages.

Page	What you do here
Dataset	Import a CSV and inspect rows, statistics, distributions and correlations.
Prepare data	Describe, clean, encode, normalize and reduce the dataset with PCA.
Regression	Predict a numeric outcome, such as y1 in this example.
Classification	Predict a class from labeled examples.
Clustering	Discover groups without choosing a target column.
Compare models	Review recorded regression and classification runs.
Visualize	Explore selected columns with charts and optional ascending X sorting.

Working across pages

You can move between pages without clearing the workspace. **Reset page** clears results on that page only. On Prepare data it clears the reports, not the prepared dataset used by other pages. Importing another dataset replaces the active data; save any models or results you need first.

02 GETTING STARTED

Import and inspect your dataset

A feature is an input used by a model. A target is the value you want to predict. Here, X1 to X8 are features and y1 is the regression target.

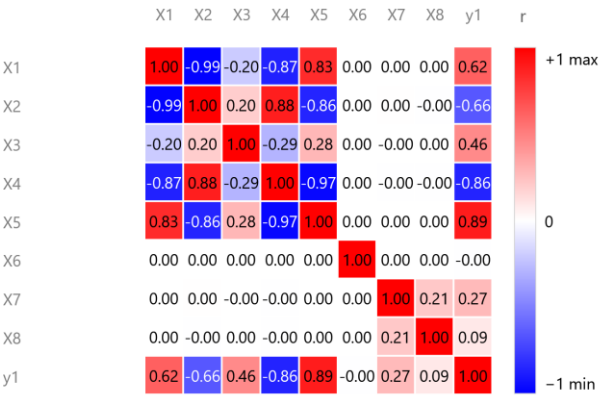
1 Prepare the CSV Use a header row, comma-separated columns and a decimal point for numeric values. Keep the same number of columns in each row. For PCA with a target, place the target last.

2 Import and check Select **Dataset**, then **Import dataset**. Check the filename, row count and column names. The supplied example shows 768 rows and 9 columns. Check missing values before training.



Figure 2 The supplied energy2.csv example after import. Confirm the displayed counts before continuing.

Read the correlation matrix



Red indicates positive correlation; blue indicates negative correlation. White is close to zero. The color ribbon runs from −1 to +1.

Here, X5 and y1 have a correlation of about 0.89. X1 and X2 are almost perfectly negatively correlated at −0.99. Strongly related inputs may contain overlapping information.

Correlation describes linear association, not causation. A value near zero does not rule out a nonlinear relationship. Use column statistics and the histogram to check ranges and unusual values too.

Figure 3 Correlations for the supplied dataset.

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Prepare features and apply PCA

Open **Prepare data** after checking the imported dataset. Choose the steps that the data needs. Cleaning and transformations change the active dataset used for subsequent training.

Tool	How to use it
Describe	Select Run Describe to review statistics for every column.
Missing Values	Run Detect Missing. Choose Mean, Max, Min, Mode or Remove Row, then Handle Missing. Check that the choice fits your data.
Anomaly Detection	Review flagged rows before enabling Remove anomalies from data. An unusual observation can still be valid.
Normalize	Choose Z-Score, Min-Max or Max-Abs to rescale columns. Keep the original data and its units documented.
Label Encoder	Convert categorical values to numeric labels. Keep the mapping; numeric codes do not automatically imply an ordered measurement.
PCA	Replace correlated inputs with fewer components. Preserve the target when preparing regression or classification data.

Choose how many PCA components to retain

1 Preserve the target For this example, y1 must be last. Keep **Last column is target (Y)** checked so PCA transforms only X1 to X8.

2 Choose the mode Select **Variance Threshold** and enter **0.95** to retain at least 95% of feature variance, or choose **Fixed Components** and enter a count. Then select **Run PCA**.

3 Check the result Review the components retained and the cumulative variance chart. A 95% variance threshold is not 95% prediction accuracy; evaluate the trained model separately.

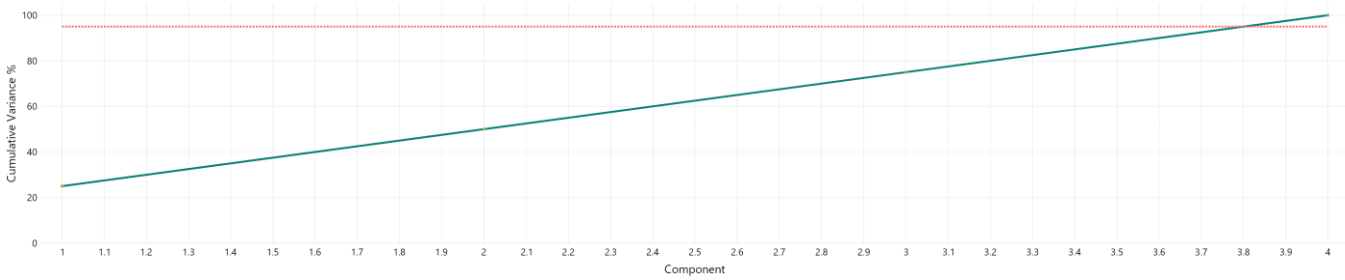


Figure 4 Components retained (horizontal) versus cumulative variance in percent (vertical). The red line marks 95%.

Enter original features when predicting

After training, the prediction form asks for original feature values. The model applies its saved normalization and PCA steps internally and returns the target in its original units. Save the model to retain this recipe.

Validation note Normalize and PCA are applied before the training split in the current app. K-fold validation does not refit these steps within each fold. Treat these scores as exploratory; independent evaluation is needed for a strict estimate of performance on new data.

04 GETTING STARTED

Train and evaluate a regression model

Use regression when the target is numeric. Linear Regression provides a baseline before trying more flexible methods. The supplied chart illustrates how to read a result; it is not a guaranteed outcome from the settings below.

1 Select the model Open **Regression**, choose **Linear Regression** and select **y1** as the **Target Column**. After PCA, the model uses the retained components internally.

2 Set up the first run Use an **80/20** train/test split with **Use K-Fold CV** off. Start with the displayed iteration and learning-rate values and **Regularization: None**. Select **Train model**.

3 Inspect the fit Check the loss curve and held-out test results. If loss rises or becomes unstable, lower the learning rate. If it is still falling, consider more iterations. Compare test performance after each change.

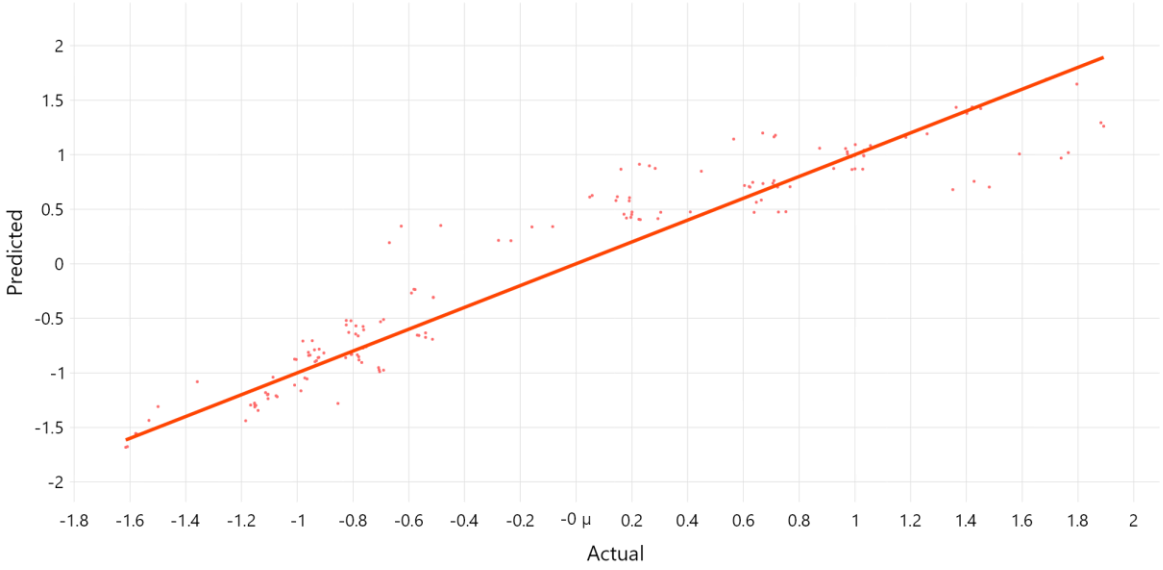


Figure 5 Supplied linear regression test scatter. Points near the diagonal indicate predictions close to actual values.

Metric	How to interpret it
R^2	Higher is better; 1 is a perfect fit. Negative values can occur. The linear summary uses a percentage scale; the fold table uses ordinary R^2.
MAE and RMSE	Lower is better. RMSE penalizes larger errors more strongly. If the target was normalized, training metrics remain in the transformed target scale.

Validate and predict

For a follow-up run, enable **Use K-Fold CV**, enter **5** folds and train again. The fold table shows training and test scores. **Combined** pools held-out predictions, with each row tested once; it is not the score of the final model fitted to all rows.

In **Predict**, enter original feature values and select **Predict**. Use **Save Model** to keep the model and preprocessing recipe. **Save tables (CSV)** exports numbers; **Save charts / images** exports plots; **Save page image** captures the current visible page.